



Fig. 2: PDH multiplexing hierarchies used in the different geographical areas

The basic digital multiplexing standard established in the United States, called the Bell System Level 1 PCM Standard, or the Bell T1 Standard, multiplexes 24 separate voice channels together, resulting in a primary rate of 1544 kbit/s. This basic rate is also being followed by Canada and Japan. In the USA's (North American) DS1 System each voice channel is 64 kbps and is designated as digital signalling Level 0, or DS-0. Each frame in the 24-channel multiplexer consists of 8 bits/channel $\times$ 24 channels + 1 framing bit = 193 bits. The total data rate when transmitting 24 channels is determined by 193 bits/frame $\times$ 8000 frames/s = 1.544 Mbps = T1, or DS-1 designation.

If four T1 lines are multiplexed together, we get 4×24 channels = 96 channels = 6.312 Mbps = T2, or DS-2 designation. Multiplexing seven T2 lines together we get 7×96 channels = 672 channels = 44.736 Mbps = T3, or DS-3 designation. If six T3 lines are multiplexed together, we get 6×672 = 4032 channels = 274.176 Mbps = T4, or DS-4 designation.

Similar hierarchy has also been adopted by Japan, but with slight deviations at higher order multiplexing/de-multiplexing levels. Japanese PDH system multiplexes 24 channels together, which results in basic rate of

1544 kbps. At the second order, four such lines are multiplexed together. Taking some framing bits, we get 4×24 channels = 96 channels = 6.312 Mbps. At the third order level, five second order lines are multiplexed together resulting in 5×96 channels = 480 channels = 32.064 Mbps. Again, multiplexing three such third order lines results in 3×480 channels = 1440 channels = 97.728 Mbps. Multiplexing such four fourth order lines results in bit rate = 4×1440 channels = 5760 channels = 397.2 Mbps.

General structure of PDH signals

The PDH signal is a serial signal stream with a frame structure formed by bit-interleaving the various signals carried within its structure. A general frame structure for a 2.048 Mbps bit stream is shown in Fig. 3. Each frame consists of 30 channels in channel-associated signalling (CAS) scheme or 31 channels in common-channel signalling (CCS) scheme. Sampling rate of each channel is 8000 samples/s (frame duration 125 microseconds) and there are 8 bits/sample. Thus, the basic speed of each channel is 8000 samples/s $\times$ 8 bits/sample = 64 kbps. In total, there are 32 time slots in each frame (designated as TS0, TS1, ..., TS31)

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